EKITI STATE UNIVERSITY, ADO -EKITI, NIGERIA
FACULTY OF SCIENCE
DEPARTMENT OF CHEMISTRY
SECOND SEMESTER B.Sc. EXAMINATIONS
2018/2019 SESSION
NOVEMBER/DECEMBER, 2019

Course Code: CHM 102 (Introductory Chemistry II) **Units** (4) **Time** 1 ½ Hrs **Question Type:** B

Instruction: *Answer all questions and shade the appropriate box*

MATRIC /REG. NO: ..... DEPT: ..... FACULTY:.....

1	A	B	C	D	21	A	B	C	D	41	A	B	C	D	61	A	B	C	D
2	A	B	C	D	22	A	B	C	D	42	A	B	C	D	62	A	B	C	D
3	A	B	C	D	23	A	B	C	D	43	A	B	C	D	63	A	B	C	D
4	A	B	C	D	24	A	B	C	D	44	A	B	C	D	64	A	B	C	D
5	A	B	C	D	25	A	B	C	D	45	A	B	C	D	65	A	B	C	D
6	A	B	C	D	26	A	B	C	D	46	A	B	C	D	66	A	B	C	D
7	A	B	C	D	27	A	B	C	D	47	A	B	C	D	67	A	B	C	D
8	A	B	C	D	28	A	B	C	D	48	A	B	C	D	68	A	B	C	D
9	A	B	C	D	29	A	B	C	D	49	A	B	C	D	69	A	B	C	D
10	A	B	C	D	30	A	B	C	D	50	A	B	C	D	70	A	B	C	D
11	A	B	C	D	31	A	B	C	D	51	A	B	C	D	71	A	B	C	D
12	A	B	C	D	32	A	B	C	D	52	A	B	C	D	72	A	B	C	D
13	A	B	C	D	33	A	B	C	D	53	A	B	C	D	73	A	B	C	D
14	A	B	C	D	34	A	B	C	D	54	A	B	C	D	74	A	B	C	D
15	A	B	C	D	35	A	B	C	D	55	A	B	C	D	75	A	B	C	D
16	A	B	C	D	36	A	B	C	D	56	A	B	C	D	76	A	B	C	D
17	A	B	C	D	37	A	B	C	D	57	A	B	C	D	77	A	B	C	D
18	A	B	C	D	38	A	B	C	D	58	A	B	C	D	78	A	B	C	D
19	A	B	C	D	39	A	B	C	D	59	A	B	C	D	79	A	B	C	D
20	A	B	C	D	40	A	B	C	D	60	A	B	C	D	80	A	B	C	D

1. The major difference between carbon and other group IVA elements include the following except
(A). Carbon has a smaller size that the rest members (B). Carbon has high electronegativity value and no d-orbital electron (C). Carbon is limited to a coordination number of four (D). Carbon has no unique ability to form multiple bonds
2. An atom X has atomic number of 56 and atomic mass of 135. The nucleus of X then has
(A). 56 protons and 135 electrons (B). 79 protons and 56 electrons (C). 79 protons and 56 neutrons
(D). None of these
3. Each of X^+ and Y^- has a total of 18 electrons. The nuclei of the neutral atoms of X and Y respectively contain
(A). 20 and 20 protons (B). 19 and 19 protons (C) 17 and 19 protons (D). 19 and 21 protons
4. Pauli exclusion principle deals with (A). collision theory of reaction rates (B). electronegativity values of elements (C). quantum numbers of orbitals/electrons (D). reversibility of equilibrium reactions
5. The atomic radius of Be, Mg and Ca are 1.13°A, 1.60°A and 1.97°A respectively. This gradation in atomic radius is due to the fact that (A). electronegativity decrease from Be to Mg to Ca (B). metallicity decreases from Be to Mg to Ca (C). the elements are all in the same period (D). the number of electronic shells increases from Be to Mg to Ca
6. Which of the following arrangements is in the order of decreasing electropositivity
(A). Fluorine, Boron, Berylium, Nitrogen, Lithium (B). Lithium, Berylium, Boron, Nitrogen, Fluorine
(C). Fluorine, Nitrogen, Boron, Berylium, Lithium (D). Lithium, Nitrogen, Boron, Fluorine, Berylium

7. All the following properties decrease down a group in the periodic table except
(A) electron affinity (B) atomic radius (C). ionization energy (D). electronegativity
8. Elements in the same period of the periodic table show
(A) similarities in chemical properties (B). progressive increase in atomic number
(C) progressive decrease in electronegativity (D). similarities in electronic structures
9. The lower the electronegativity difference between two elements, the
(A) higher the polarity of their interatomic bond (B). lower the polarity of their interatomic bond
(C) weaker their interatomic bond (D). higher the possibility of forming a molecule
10. Which of the following halogens has the lowest electropositivity (A). Cl (B). I (C). F (D). Br
11. If 12 and 9 are the atomic numbers of elements A and B respectively, the interatomic bond between them is.....
(A). Dative (B). Covalent (C). Ionic (D) Metallic
12. All the following statements are false about NaCl in the solid state except...
(A). It exists as aggregates of ions. (B). It conducts electric current
(C). It exists as discreet molecules (D). Its ions are linked by metallic bonds.
13. When the Lithium atom forms its ion,
(A). It's atomic number decreases (B). It attains the electronic configuration of helium
(C). It gains an extra electron and becomes negatively charged. (D). It loses one proton.
14. If the electronegativity difference between element A and B is small, the interatomic bond between them is...
(A). dative (B). polar covalent (C). ionic (D). non polar covalent
15. A molecule of Nitrogen contain------bond (A). double (B). triple (C). single (D). multiple
16. The bonds that hold components of metallic, ionic network and molecular solids together are
(A). sea of electrons, electrovalent forces, intermolecular forces, covalent forces respectively
(B). sea of electrons, covalent forces, electrovalent forces, intermolecular forces respectively
(C). sea of electrons, electrovalent forces, covalent forces, electrovalent forces respectively
(D). sea of electrons, covalent forces, intermolecular forces, electrovalent forces respectively
17. Which of the following classification is incorrect (A). $H_2O_{(s)}$, molecular solid (B). $KF_{(s)}$, Ionic solid
(C). $SiC_{(s)}$, covalent solid (D). $S_{(s)}$, metallic solid
18. Be, Cu, Zn and Au adopt which packing methods? (A) HCP, CCP, HCO, CCO respectively (B). CCP, HCP, HCP, CCP respectively (C). HCP, HCP, CCP, CCP respectively (D). CCP, CCP, HCP, CCP respectively
19. A unit cell that contain lattice points only at the corners is known as what? How many lattice point does such unit cell contain? (A). face centered unit. Contains four lattice points (B). primitive unit cell. Contains one lattice point (C). body centered unit cell. Contains two lattice points (D). side centered unit cell. Contains two lattice points
20. The crystal system is a format by which crystal structures are classified and each crystal system is defined by the relationship between edge lengths a, b and c. which axial relationship is representative for monoclinic, orthorhombic and triclinic systems (A). $a \neq b = c$ (B). $a \neq b \neq c$ (C). $a = b \neq c$ (D). $a = b = c$
21. Given that atomic radius is defined as half the distance between the centres of two neighbouring atoms, what is the atomic radius of BCC crystal structures if the unit cell length is "a" (A). a/2 (B). a/4 (C). $a/4\sqrt{2}$ (D). $a/4\sqrt{3}$
22. Xenon crystallizes in the face-centered cubic lattice and the edge of the unit cell is 620 pm. What is the distance between the nearest neighbours (A). 620 pm (B). 219.23 pm (C). 438.47 pm (D). 525 pm
23. Two ions A^+ and B^- have radii 88 and 200 pm respectively. In the closed packed crystal of compound AB, predict the coordination number of A (A). 4 (B). 8 (C). 6 (D). 12
24. A unit cell of Au has a Au at coordinates 0, 0, 0 and another Au at coordinates 0, ½, ½; ½, 0, ½; ½, ½, 0. How many atoms of Au are there in the unit cell (A). 4 (B) 8 (C). 6 (D). 12
25. Close packing is maximum in the crystal that is —— (A). Simple cubic (B). BCC (C). FCC (D). None of these
26. Solvent extraction is a separation technique which is based on the principle of
(A). Distribution of a solute between two immiscible solvents
(B). Distribution of a solvent between two immiscible solvents
(C). Distribution of a solute between two miscible solvents
(D). Distribution of a solvent between two miscible solutes
27. The purification of liquid systems based on differences in the boiling points of the components is called
(A). Substitution (B). Distillation (C). Distribution (D). Separation
28. Technique based on the distribution of substances between stationary phase and mobile phase is called
(A). Eluent Chromatography (B). Gas Chromatography (C). Column Chromatography (D) Moving

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